Transfer task: plan and begin

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Organize the given measurements and details. Begin your own position prediction.

SUCCESS CRITERIA - I can...

1. Organize the task data with fixed axes and matching units.
2. Calculate central and bounded final coordinates.
3. Save an explicit plan for the graph and explanation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “bound” to explain your answer. Sentence starters are optional.

component: the part of a vector along one chosen axis

bound: an upper or lower limit in a model

assumption: a condition accepted for using a model

01 Warm-up: bring back an earlier idea

A **3 kg** lamp hangs at rest from a vertical rope. Use g=**9.8**. **Find tension**. Then **find tension** if it accelerates upward at **2 m/s²**.

Word help: tension = the pulling force through a stretched rope or cable

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

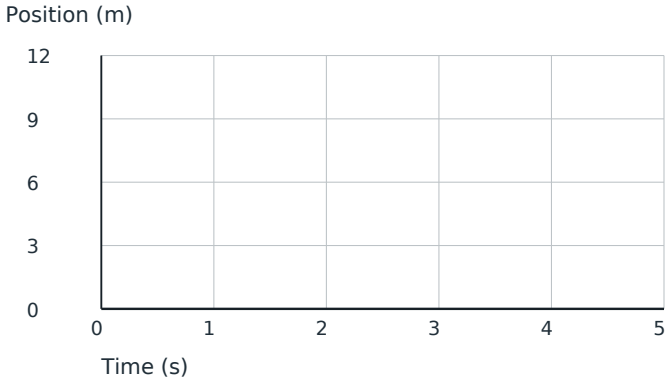
02 Worked example: explain the prediction

Reference: $v_x = v \cos\theta$; $v_y = v \sin\theta$; $x_f = x_i + v_x t$ $y_f = y_i + v_y t$.
For this fictional calculation, treat Earth as fixed: x_i = **-1.2×10^8 km**; y_i = **0**; v = **8500 m/s**; θ = **15°** from +x toward +y; t = **1.6×10^7 s**. Convert units.

Worked reasoning: A simplified calculation uses x_0 = **0 m** and constant v = **2 m/s** for **5 s**, giving x = **10 m**. This prediction depends on velocity staying constant. It does not prove that the real object moves that way.

Your turn: Draw the predicted path and **mark one observation you would** collect to **test the idea** that velocity stays constant.

Sentence starter: I predict ... . I used ... to make this prediction because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Plan and represent

A simplified prediction of a fictional asteroid’s path fixes Earth at the origin and ignores gravity and Earth's motion.
1. Use the givens to plan the unit conversion, velocity components, final position, bounds and graph.
2. **Sketch fixed** +x and +y axes, with +x initially toward Earth.

Sentence starter: My design includes … . It does what is needed by … .

Quantity	Value
Initial x	-1.2×10^{8} km
Initial y	0 km
Speed	8500 m/s
Direction	15° from +x toward +y
Elapsed time	1.6×10^{7} s

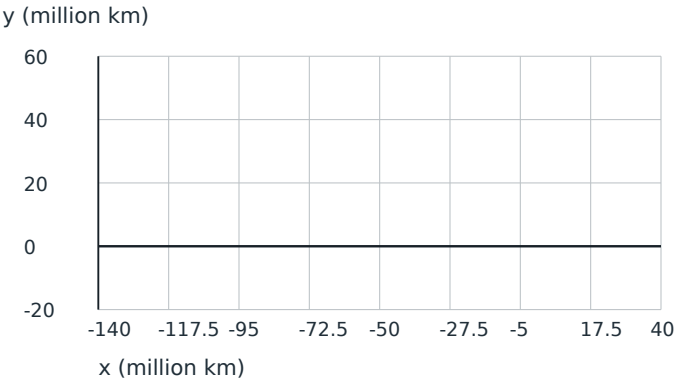
Marzano check - circle: ① ② ③

04 Central prediction

Resolve the velocity in question **3** and **calculate final x** and y in kilometers. Keep unrounded components until the end. Plot the path and distinguish final position, displacement and distance from the origin.

Word help: resolve = separate a vector into components

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Propagate bounded inputs

The initial x bound is ±**50,000 km** and the speed bound is ±**200 m/s** at the same **15°** angle. Initial x and speed vary independently; angle, initial y and time are exact. **Find final coordinate half-widths** and intervals.

Sentence starter: I found … [unit]. This value means … in this situation.

Coordinate	Half-width (km)	Lower (km)	Upper (km)
x	___	___	___
y	___	___	___

Marzano check - circle: ① ② ③

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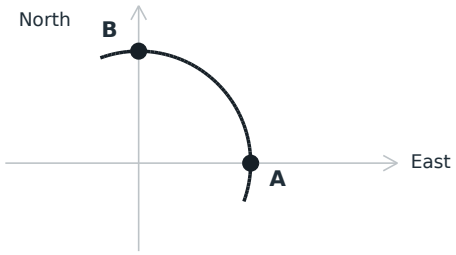
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A puck follows a circle at a steady **7 m/s**. A student calls it equilibrium. **Draw velocity** at two points and correct the statement.
Use the curve for the turning case; **show how velocity changes**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Path sketch: runner moves from A toward B.

Marzano check - circle: (1) (2) (3)

07 Checkpoint

Prepare the central, lower and upper x values at the start and final time for tomorrow's graph.
1. Use questions **3**-**5**.
2. **State one thing you** are treating as true, one omitted effect, and the next step you need to finish.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Time	Central x	Lower x	Upper x
Start	____	____	____
Finish	____	____	____

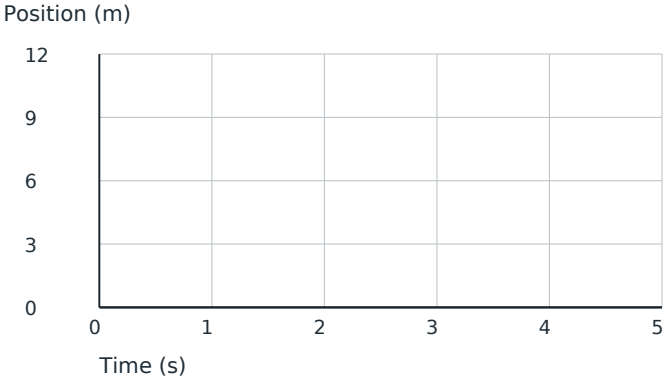
Marzano check - circle: (1) (2) (3)

08 STRETCH: test the prediction

A calculation predicts x = **10 m**, but the measurement is **8.0 ± 0.2 m**.
1. Create **two possible descriptions** of a **5 s** trip to **8 m**: one with constant velocity, and one starting from rest with constant acceleration. Include calculations.
2. **Explain what extra measurements could** help you choose between them.

Word help: resolve = separate a vector into components

Sentence starter: I predict ... . I used ... to make this prediction because ... .



Marzano check - circle: (1) (2) (3)